

university of groningen

Performance Engineering: Lecture #05

WMCS045-05

Dirk Pleiter

Bernoulli Institute

2026-09-14

Overview



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

Content



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

Semi-Analytical Performance Modelling



Procedure: [Hoefler, 2011]

- **Analytic steps**
  - Identify application kernels
  - Identify input parameters that influence runtime
    ☞ Parametrization of work-load W
  - On basis of prior knowledge formulate scaling formulae describing dependence of latency $\Delta t(W)$ as function of work-load W
- **Empirical steps**
  - Measure $\Delta t(W)$ for different W
  - Fit scaling formulae to measurements
- **Model verification**
  - Compare predicted and measured values
- **Parameter validation**
  - Check fitted parameters for plausibility

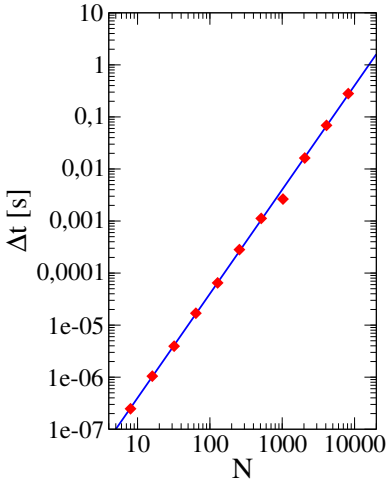
Example (1/2)

- **Kernel**: Matrix-vector multiplication

$$y_i \leftarrow \sum_{j=0}^{N-1} A_{ij} x_j \quad (i = 0, \ldots, N-1)$$

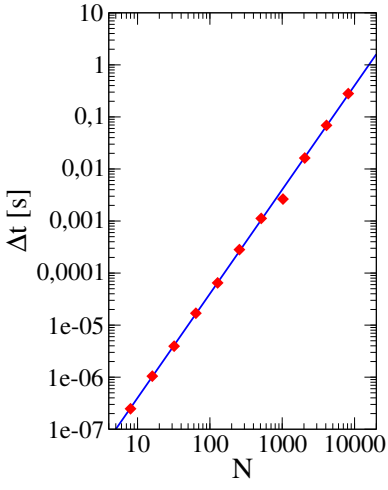
- **Ansatz**

$$\Delta t(N) = a_0 + a_2 N^2$$



Example (2/2)

- **Test setup**:
  - Double-precision floating-point numbers
  - Intel i7-2620M processor, single thread
  - No compiler optimization
- **Results**:
  - $a_0 \simeq 0\,\text{ns}$, $a_2 \simeq 4\,\text{ns}$
- **Plausibility analysis**
  - Clock $f = 2.7\,\text{GHz} \Rightarrow 1/f = 0.4\,\text{ns}$
  - Need about 10 cycles per (i, j)



Content



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

Information Exchange Function



- Each task on the computer eventually implies that information is transferred from a storage device x to a storage device y
  - Machine model = abstract machine model
- **Information Exchange Function**:

$$I^k_{x,y}(W) \quad = \quad \text{data transferred between computer sub-systems for specific computational task } k$$

x ... source storage device (e.g. memory)
y ... destination storage device (e.g. register file)
W ... workload

Bandwidth or Throughput vs. Latency (1/2)



Reminder:

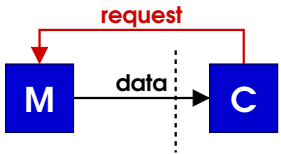
- Definitions bandwidth versus throughput:
  - **Bandwidth** = Amount of items passing a particular interface per time unit
  - **Throughput** = Amount of work done in a given time
- Definition **latency (or response time)** $\Delta t =$
  Time between events indicating start of an task and the event signalling its completion

Bandwidth or Throughput vs. Latency (2/2)



Example:

- Memory read operation

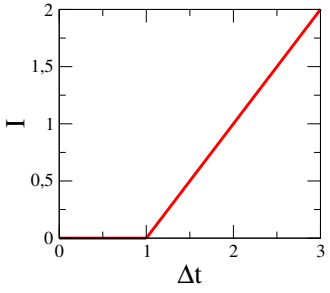


- **Bandwidth**: Amount of data passing memory interface per time unit
- **Latency**:
  - **Start event**: Memory read operation started
  - **End event**: All data arrived in processor
- **Note**: This is an abstract model and details remain open
  - Example: Load instruction versus cache line read command

Latency-Bandwidth Model (1/2)

- Simple model describing **latency** Δt as a function of the information exchange I
- Model parameters:
  - **Start-up latency** λ = Time between event trigger and start of arrival of response
  - **Bandwidth** β: Parametrizes time to exchange information I
- Model:

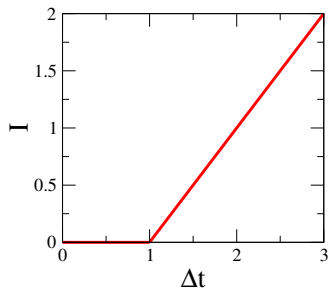
$$\Delta t(I) = \lambda + \frac{I}{\beta}$$

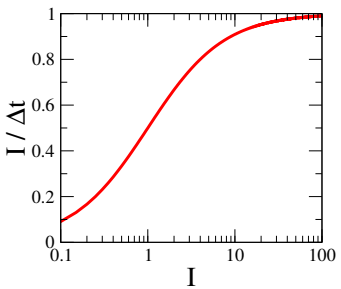


Latency-Bandwidth Model (2/2)

- **Effective bandwidth** $\tilde{b} = I/\Delta t(I)$
- Example: Read I Bytes from memory or network

$$\Delta t(I) = \lambda + \frac{I}{\beta} \quad \Rightarrow \quad \tilde{b}(I) = \left(\frac{\lambda}{I} + \frac{1}{\beta}\right)^{-1}$$





Information Exchange Model: Latency Predictions

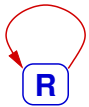


Ansatz to predict latency:

$$\Delta t^k_{x,y} \simeq \lambda_{x,y} + \frac{I^k_{x,y}}{\beta_{x,y}}$$

Example:

- x, y = R
- $\beta_{R,R}$ = throughput arithmetic unit
- $I^k_{x,y}$ = number of input (or output) operands



arithmetic unit

register file

Beware of limitations of this ansatz:

- Transfer mechanism may depend on task size N
- Bandwidth changes due to resource congestion is ignored
- ...

Simple Example (1/2)

- Example operation: $y_i \leftarrow \alpha \cdot x_i$ $(i = 1, \ldots, N)$
  x_i, y_i ... vectors of single precision floating-point numbers
  α ... single-precision floating-point number
- Information exchange for each i:

load x_i	4 Bytes
multiply α and x_i	1 Flop
store y_i	4 Bytes

- Assume the following hardware parameters:

Memory bandwidth β_{mem}	1 Byte/clock cycle
Floating-point unit throughput β_{fp}	1 Flop/clock cycle

- Latency predictions:

Δt_{mem}	$N \cdot 8$ clock cycles
Δt_{fp}	$N \cdot 1$ clock cycles

Simple Example (2/2)



- **Latency** for full operation:
  - No overlap of memory load/store and arithmetic operations:

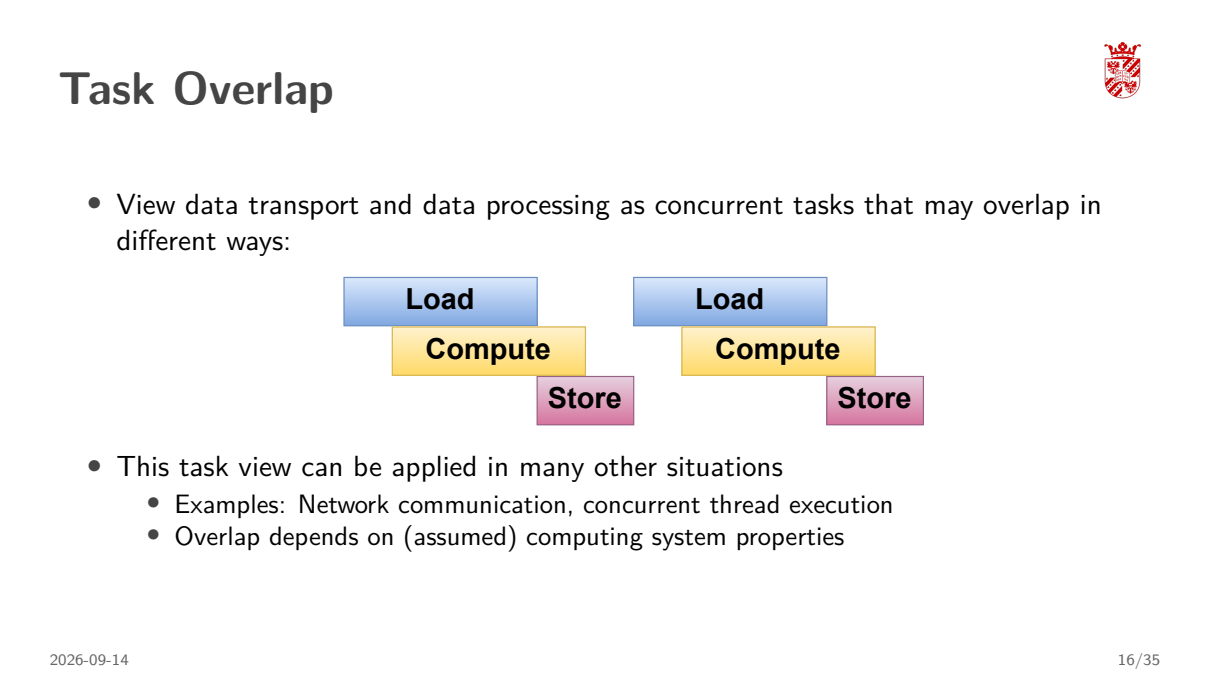
$$\Delta t(N) = \Delta t_{\text{mem}} + \Delta t_{\text{fp}} = N \cdot 9$$

  - Perfect overlap of memory load/store and arithmetic operations:

$$\Delta t(N) = \max\left(\Delta t_{\text{mem}}, \Delta t_{\text{fp}}\right) = N \cdot 8$$

☞ **Memory (bandwidth) limited** problem
  - Opposed to **compute (throughput) limited** problems

Task Overlap



- View data transport and data processing as concurrent tasks that may overlap in different ways:

- This task view can be applied in many other situations
  - Examples: Network communication, concurrent thread execution
  - Overlap depends on (assumed) computing system properties

Memory Bandwidth



- Definitions
  - Measured bandwidth b_{mem}
    - Amount of transferred data I_{mem} over measured latency Δt
  - Latency-bandwidth model parameter β_{mem}
  - Effective bandwidth $\tilde{b}_{\mathrm{mem}}$
  - **Nominal hardware bandwidth** B_{mem}
    - Data rate on hardware link/bus
- Relations
  - Upper limit for measured bandwidth: $b_{\mathrm{mem}} \leq B_{\mathrm{mem}}$
  - Upper limit for effective bandwidth: $\tilde{b}_{\mathrm{mem}} \leq \beta_{\mathrm{mem}}$
  - Note: $\beta_{\mathrm{mem}} \leq B_{\mathrm{mem}}$ is not guaranteed
    - But expected if bandwidth-latency model is a good model

Semi-Analytical and Information Exchange Modelling

- **Ansatz**: Use Information Exchange to create semi-analytical ansatz, e.g.

$$\Delta t = a_0 + a_1 I(W)$$

  - a_1 has dimension of inverse throughput/bandwidth
- **Advantages**
  - No need to determine start-up latency λ and asymptotic bandwidth β
  - Allow for easy comparison with hardware performance parameters

Example



- Double-precision matrix-vector multiplication

$$y_i \leftarrow \sum_{j=0}^{N-1} A_{ij}\, x_j \quad (i = 0, \ldots, N-1)$$

- Information exchange

$$\begin{aligned} I_{\mathrm{mem,rf}} &= (N^2 + 2\,N)\,8\,\mathrm{B} \\ I_{\mathrm{rf,rf}} &= (2\,N - 1)\,N\,\mathrm{Flop} \end{aligned}$$

- Re-use previously presented results for Intel i7-2620M and comparison to nominal throughput/bandwidth B

Storage devices	$1/a_1$	B
mem,rf	$2\,\mathrm{GB/s}$	$21.3\,\mathrm{GB/s}$
rf,rf	$0.5\,\mathrm{GFlop/s}$	$43.2\,\mathrm{GFlop/s}$

Content



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

Arithmetic Intensity



- **Arithmetic Intensity** $=$ [H. Harris, 2005]

$$\mathrm{AI} = \frac{\text{Number of floating-point operations}}{\text{Amount of transferred data}} = \frac{I_{\mathrm{fp}}}{I_{\mathrm{mem}}}$$

- Example: $y_i \leftarrow \alpha \cdot x_i$

$$\mathrm{AI} = \frac{1\,\mathrm{Flop}}{8\,\mathrm{B}} \qquad \text{single-precision}$$

$$\mathrm{AI} = \frac{1\,\mathrm{Flop}}{16\,\mathrm{B}} \qquad \text{double-precision}$$

Roofline Model (1/2)



[S. Williams et al.; 2009]

- Floating-point performance

$$b_{\mathrm{fp}} = \frac{I_{\mathrm{fp}}}{\Delta t} \quad \Rightarrow \quad \Delta t = \frac{I_{\mathrm{fp}}}{b_{\mathrm{fp}}}$$

- Floating-point and memory performance limits:

$$b_{\mathrm{fp}} \leq B_{\mathrm{fp}}, \quad b_{\mathrm{mem}} \leq B_{\mathrm{mem}}$$

- Upper limit for latency assuming perfect overlap of memory and arithmetic operations:

$$\Delta t \geq \max\left(\Delta t_{\mathrm{fp}}, \Delta t_{\mathrm{mem}}\right)$$

Roofline Model (2/2)



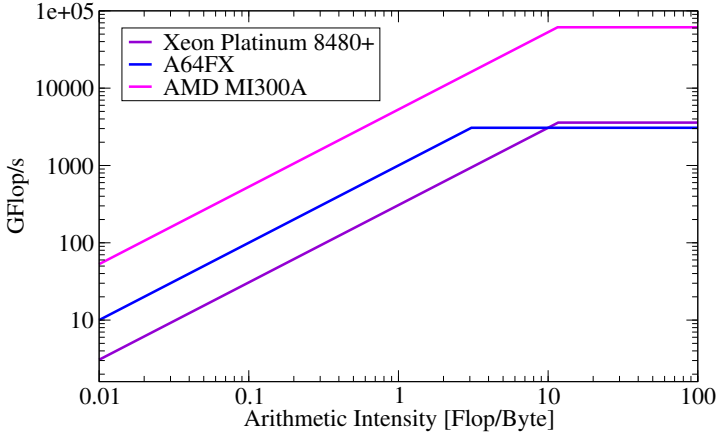
- Assume latency-bandwidth model, but neglect start-up latencies

$$\Delta t \geq \max\left(\lambda_{\mathrm{fp}} + \frac{I_{\mathrm{fp}}}{B_{\mathrm{fp}}}, \lambda_{\mathrm{mem}} + \frac{I_{\mathrm{mem}}}{B_{\mathrm{mem}}}\right) \quad \Rightarrow \quad \frac{1}{\Delta t} \leq \min\left(\frac{B_{\mathrm{fp}}}{I_{\mathrm{fp}}}, \frac{B_{\mathrm{mem}}}{I_{\mathrm{mem}}}\right)$$

- Upper limit (=roof) for floating-point performance ("attainable performance")

$$b_{\mathrm{fp}} \leq \min\left(B_{\mathrm{fp}}, \frac{I_{\mathrm{fp}}}{I_{\mathrm{mem}}} B_{\mathrm{mem}}\right) = \min\left(B_{\mathrm{fp}}, \mathrm{AI} \cdot B_{\mathrm{mem}}\right)$$

Roofline Examples



Cache-Aware Roofline Models



- The naive Roofline model only considers bandwidth towards the external memory
- **Cache-aware roofline models**: Account for accesses to the different data cache levels
- **One possible approach**: Consider B_{mem} as a function of I_{mem} [Ilic et al.; 2014]
  - For sufficiently small I_{mem} set

$$B_{\mathrm{mem}}(I_{\mathrm{mem}} \ll C_{\mathrm{L}_1}) = B_{\mathrm{L}_1}$$

- **Alternative approach**: Define information exchange for the different levels
  - This results in different values for AI

Content



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

ECM Model (1/3)

[J. Treibig and G. Hager, 2010]

- **ECM** = Execution-Cache-Model
- **Goal**: Analytic model to predict performance of bandwidth-limited algorithms in all memory hierarchy levels
- **Ansatz** for overall latency

$$\Delta t = \max(\Delta t_{\mathrm{OL}}, \Delta t_{\mathrm{nOL}} + \Delta t_{\mathrm{mem}})$$

Δt_{mem}	...	Cycles spent on load/store up to L1
Δt_{OL}	...	Cycles spent on overlapping instructions
Δt_{nOL}	...	Cycles spent on non-overlapping instructions

ECM Model (2/3)



- **Ansatz** for data transport latency
  - Data resides in L1 cache

$$\Delta t_{\mathrm{mem}} = 0$$

  - Data resides in external memory

$$\Delta t_{\mathrm{mem}} = \Delta t_{\mathrm{L1,L2}} + \cdots + \Delta t_{\mathrm{LLC,mem}}$$

ECM Model (3/3)



- **Conventions**
  - All memory transfers in **units of cache lines (CL)**
  - All times to be understood as inverse bandwidth, i.e. $\Delta t_{\mathrm{mem}} = 1\,\mathrm{CL}/B_{\mathrm{mem}}$
  - Time in units of core clock cycles
  - External memory bandwidth is determined phenomenologically
- Short-cut notation for model parameters (assuming L3 = LLC)

$$\{\Delta t_{\mathrm{OL}} || \Delta t_{\mathrm{nOL}} | \Delta t_{\mathrm{L1,L2}} | \Delta t_{\mathrm{L2,L3}} | \Delta t_{\mathrm{L3,mem}}\}$$

- Short-cut for model predictions

$$\{\Delta t_{\mathrm{L1}} \rceil \Delta t_{\mathrm{L2}} \rceil \Delta t_{\mathrm{L3}} \rceil \Delta t_{\mathrm{mem}}\}$$

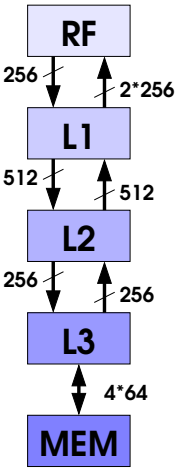
  where Δt_{L1} is the prediction if data resides in L1

Intel Xeon E5-v3 (Haswell) Memory Hierarchy



- 2 CL can be loaded to RF in 2 cycles
- Bandwidth to memory is shared by all cores
- Example memory bandwidth
  - Core clock speed $f = 2.3\,\text{GHz}$
  - $b_{\text{mem}} = 32.4\,\text{GB/second}$ (measured)
  - Cache line load latency

$$\begin{aligned}\Delta t_{\text{mem}} &= (64\,\text{B/CL}) \\ &\quad \cdot (32.4\,\text{GB/second})^{-1} \\ &\quad \cdot (2.3\,\text{GHz}) \\ &\simeq 4.54\,\text{cycle/CL}\end{aligned}$$



ECM Model Example (1/2)



[J. Hofmann et al.; 2015]

- Computational task

$$c \leftarrow \sum_i a_i \cdot b_i$$

- ECM model parameter determination
  - Processor is capable of performing 2 256-bit SIMD FMA per cycle
    - Need 1 cycle to process a full cache line
    - $\Delta t_{\mathrm{OL}} = 1$
  - Load of 2 CL from L1→RF and L2→L1 requires 2 cycle
    - $\Delta t_{\mathrm{L1,L2}} = 2$
  - Assume that loads are non-overlapping
    - $\Delta t_{\mathrm{nOL}} = 2$
  - Bandwidth from L3→L2 only 256 bit/cycle
    - $\Delta t_{\mathrm{L2,L3}} = 4$
  - Load of 2 CL from MEM requires $2 \cdot 4.54$ cycle
  - Short notation

$$\{1 \,||\, 2 \,|\, 2 \,|\, 4 \,|\, 9.1\} \text{ cycle}$$

ECM Model Example (2/2)

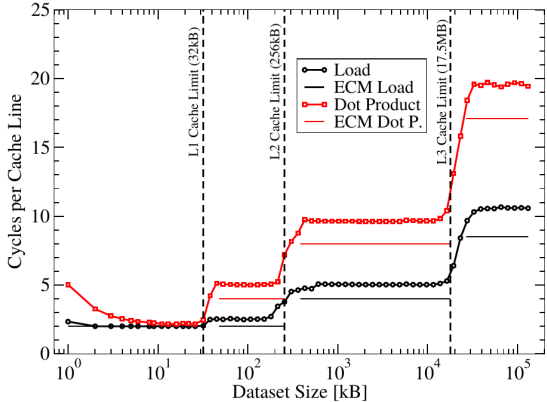
- ECM model prediction

$$\{2 \rceil 4 \rceil 8 \rceil 17.1\} \text{ cycle}$$

- Measured timings on Xeon E5-2695 v3 (Haswell, 14 cores)

$$\{2.1 \rceil 4.7 \rceil 9.6 \rceil 19.4\} \text{ cycle}$$

[J. Hofmann et al.; 2015]



Content



1. Semi-Analytical Performance Models
2. Information Exchange
3. Roofline Model
4. ECM Model
5. Final

Summary Lecture #05



- Modelling techniques
  - Semi-Analytical performance models
  - Information Exchange and Latency-Bandwidth model
  - Roofline model and cache-aware Roofline model
  - Execution-Cache-Model (ECM)

Questions?



[Wiki Media]